Python Code

```python
import os
import pypdf
import streamlit as st
import yfinance as yf
import google.generativeai as genai

# --- CONFIGURATION PAGE (Mobile Friendly) ---
st.set_page_config(
    page_title="AI Stock Analyst",
    page_icon="📈",
    layout="centered",
    initial_sidebar_state="collapsed",
)

st.title("📈 AI Stock & Industry Analyst")
st.caption(
    "Analisis Saham Real-Time, Laporan Keuangan PDF, & Prediksi Industri Masa"
    " Depan"
)

# --- SIDEBAR: API KEY CONFIGURATION ---
with st.sidebar:
  st.header("⚙️ Konfigurasi Agent")
  api_key = st.text_input(
      "Masukkan Gemini API Key:",
      type="password",
      help="Dapatkan API key gratis dari Google AI Studio",
  )

# --- HELPER FUNCTIONS ---


def get_realtime_stock_data(ticker_symbol):
  """Mengambil data pasar real-time dan rasio fundamental dari Yahoo
Finance."""
  try:
    ticker = yf.Ticker(ticker_symbol)
    info = ticker.info

    # Pastikan data esensial tersedia
    metrics = {
        "Name": info.get("longName", ticker_symbol),
        "Price": info.get(
            "currentPrice", info.get("regularMarketPrice", "N/A")
        ),
        "Currency": info.get("currency", "USD"),
        "PE_TTM": info.get("trailingPE", "N/A"),
        "Forward_PE": info.get("forwardPE", "N/A"),
        "PBV": info.get("priceToBook", "N/A"),
        "ROE": info.get("returnOnEquity", "N/A"),
```

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        "NPM": info.get("profitMargins", "N/A"),
        "DER": info.get("debtToEquity", "N/A"),
        "MarketCap": info.get("marketCap", "N/A"),
        "Sector": info.get("sector", "N/A"),
        "Industry": info.get("industry", "N/A"),
    }
    return metrics
  except Exception as e:
    return None


def extract_pdf_text(uploaded_file):
  """Membaca dan mengekstrak teks dari laporan keuangan PDF."""
  try:
    pdf_reader = pypdf.PdfReader(uploaded_file)
    extracted_text = ""
    # Batasi pembacaan halaman pertama untuk performa cepat di HP (misal 15 halaman pertama)
    max_pages = min(len(pdf_reader.pages), 15)
    for i in range(max_pages):
      extracted_text += f"\n--- Halaman {i+1} ---\n" + pdf_reader.pages[i].extract_text()
    return extracted_text
  except Exception as e:
    return f"Gagal membaca PDF: {str(e)}"


# --- MAIN INPUT FORM ---
col1, col2 = st.columns([2, 1])
with col1:
  ticker_input = (
      st.text_input("Kode Saham (Ticker):", value="NVDA")
      .upper()
      .strip()
  )
with col2:
  st.write("")
  st.write("")
  fetch_btn = st.button("🔍 Analisis", use_container_width=True)

uploaded_pdf = st.file_file_uploader(
    "Upload Laporan Keuangan (PDF, Opsional):", type=["pdf"]
)

# --- EXECUTION & ANALYSIS ---
if fetch_btn:
  if not api_key:
    st.error("Masukkan Gemini API Key terlebih dahulu di menu Sidebar (⚙️).")
  else:
    genai.configure(api_key=api_key)
    model = genai.GenerativeModel("gemini-2.5-flash")

    with st.spinner("Mengambil data real-time & memproses analisis..."):
      # 1. Fetch Real-Time Market Data
```

```
      stock_data = get_realtime_stock_data(ticker_input)

      if not stock_data:
        st.error(
            f"Kode saham '{ticker_input}' tidak ditemukan atau gagal diambil."
        )
      else:
        # Display Metric Cards (Tampilan Ringkas di HP)
        st.subheader(f"{stock_data['Name']} ({ticker_input})")
        m1, m2, m3 = st.columns(3)
        m1.metric("Harga", f"{stock_data['Price']} {stock_data['Currency']}")
        m2.metric(
            "P/E Ratio",
            f"{round(stock_data['PE_TTM'], 2) if
isinstance(stock_data['PE_TTM'], (int, float)) else 'N/A'}",
        )
        m3.metric(
            "PBV",
            f"{round(stock_data['PBV'], 2) if isinstance(stock_data['PBV'],
(int, float)) else 'N/A'}",
        )

        # 2. Extract PDF Content
        pdf_content = ""
        if uploaded_pdf:
          pdf_content = extract_pdf_text(uploaded_pdf)
          st.success("PDF berhasil dibaca.")

        # 3. Construct Agent Prompt
        agent_prompt = f"""
                Anda adalah Analis Saham & Penasihat Keuangan Profesional.
                Tugas Anda: Analisis saham {ticker_input} berdasarkan 3 sumber
data:

                DATA MARKET REAL-TIME:
                {stock_data}

                DOKUMEN LAPORAN KEUANGAN (PDF EXTRACT):
                {pdf_content if pdf_content else "Tidak ada PDF diunggah.
Gunakan estimasi fundamental & historis dasar."}

                KERANGKA ANALISIS (Wajib digunakan):
                1. Valuasi: Evaluasi P/E Ratio & PBV.
                2. Profitabilitas & Kesehatan Finansial: Bahas NPM, ROE, dan
tingkat utang (DER).
                3. Prospek Industri Masa Depan: Analisis tren teknologi,
geopolitik, dan posisi daya saing (Moat) perusahaan di industri masa depan.
                4. Kesimpulan & Rekomendasi: Berikan opsi tindakan logis
(Buy/Hold/Sell) untuk investor jangka panjang.

                Format jawaban dalam Bahasa Indonesia, terstruktur ringkas,
dan mudah dibaca di layar HP.
                """

```

```
        # 4. Generate AI Analysis
        response = model.generate_content(agent_prompt)

        st.markdown("---")
        st.subheader("📋 Laporan Analisis AI Agent")
        st.markdown(response.text)

        st.caption(
            "⚠️ Disclaimer: Seluruh hasil analisis ini bersifat edukasi dan"
            " informasi semata, bukan merupakan ajakan mutlak untuk membeli"
            " atau menjual saham tertentu."
        )

```